Quantum Benchmarks

Hardware Runs

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This benchmarking report was generated using Classiq v1.10.1.

The report includes hardware runs for a collection of pre-defined and user-defined functions and algorithms, where each benchmark can be instantiated at different problem sizes. For each benchmark experiment, we report a score defined specifically for that example, together with the execution time[1] and quantum-program properties obtained through Classiq's hardware-aware synthesis[2]. For further information, see the Benchmarking directory in classiq-library.

[1] Currently, the reported time includes both execution time and waiting time.
[2] The properties of the actual circuit executed on the hardware may differ slightly due to pre-run transpilation.

Adder

We benchmark an out-of-place addition by a constant,

$$|x\rangle_n|0\rangle_n \to |x\rangle_n|x + c \bmod 2^n\rangle_n,$$

where n is a the variable size and c is a fixed integer. In the benchmark, the input variable is prepared in a uniform superposition, and the constant is chosen as

$$c = 2^{\lfloor n/2 \rfloor} - 1.$$

The score is defined as the total measured probability of obtaining the correct arithmetic relation,

$$\text{Score} = \sum_{x,y=0}^{2^n-1} P(x, y)\, \delta_{y,\, x+c \bmod 2^n},$$

where $P(x, y)$ is the measured joint output distribution of the two variables. Equivalently, the score is the total probability mass on outcomes satisfying

$$y = x + c \bmod 2^n.$$

A score of 1 corresponds to perfect agreement with the ideal adder transformation.

For more details see Benchmark Adder notebook.

adder - 4 qubits (1000 shots)

Provider	Backend Name	Score	Time Elapsed (min)	Width	Depth	2Q Gate Count
Classiq	simulator	1.0	0.0	9	102	84
Classiq	simulator_statevector	1.0	0.0	–	–	–